

A REVIEW OF HISTORICAL AND CURRENT CODING STANDARDS FOR OPTICAL COMMUNICATION

HIDEKI HILL

ABSTRACT. Fibre optic has been a standard part of the global communication backbone for the last 40 years, and has become a household standard for consumer applications in the last 20 years. This paper gives an overview of the fibre optic channel, and the historical evolution of constraints and forward-error-correcting coding schemes. Some topics include Reed-Solomon, Turbo, and Low-Density Parity-Check codes. The paper concludes with a high-level overview of Staircase codes as a modern development in high-throughput applications. The proposal is written in this format to demonstrate the final paper's proposed template and aesthetic. Any feedback on the format, template, or content is welcome.

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1. INTRODUCTION

This section gives background on optical communication channel characteristics and applications, including the widespread adoption to consumer and commercial internet, long haul and intercontinental communication, and outer space communication. To set up proceeding comparisons between the coding schemes in following sections, an explanation of hard- and soft-decoding will be provided. Coding gain, decoding complexity, and data rate will be presented as metrics on the quality, reliability, and speed of different codes. One general reference for the paper will be the the book chapter by Benedetto and Bosco [1].

2. REED-SOLOMON CODES

This section introduces the origins (1970's to 1980's) of Reed-Solomon codes in optical channels while building upon course material. As discussed in lecture, RS codes remain relevant in modern coding theory; they were well-studied even during the initial development of the optical channel and were thus an obvious candidate for optical coding due their structured algebraic construction and existing decoding algorithms [6] [1].

3. TURBO CODES

By the 1990's, iterative decoding had a renewed interest given Berrou's invention of Turbo codes [2] and their near-Shannon limit. This section gives a technical description of a Turbo code and explains why they are better suited to optical channels than RS codes by comparing bit error rate, gain, and other factors [5]. LDPC codes had become an active research area again, and would soon supersede Turbo codes for certain applications.

4. LOW-DENSITY PARITY-CHECK CODES

This section gives a technical description of Gallager's original LDPC codes [4] and why they remain standard in many applications today, offering better performance than Turbo codes for high rate channels [1] [3]. If LDPC codes are discussed toward the end of the course, class lectures will be referenced too.

5. STAIRCASE CODES

This section gives a high-level overview of Staircase codes and how they achieve high performance at low computational complexity, thus low power draw. They will be presented as an improvement upon LDPC codes for applications to modern high-throughput optical channels as in [7].

REFERENCES

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